

Chip Basics: Time, Area, Power, Reliability, and Configurability

INTRODUCTION

- The trade-off between cost and performance is fundamental to any system design.
- The driving force behind design innovation is rapid technological advancement.
- The Semiconductor Industry Association (SIA) regularly publishes projections, called the SIA roadmap, of technology advances, which serve as the basis and assumptions for new chip designs.

TABLE 2.1 Technology Roadmap Projections

Year	2010	2013	2016
Technology generation (nm)	45	32	22
Wafer size, diameter (cm)	30	45	45
Defect density (per cm ²)	0.14	0.14	0.14
μ P die size (cm ²)	1.9	2.6	2.6
Chip frequency (GHz)	5.9	7.3	9.2
Million transistors per square centimeter	1203	3403	6806
Max power (W) high performance	146	149	130

Design Trade - Offs

- With increases in chip frequency and, especially, transistor density, the designer must find the best set of trade-offs in an environment of rapidly changing technology.
- When making basic design trade-offs, we consider five factors.
- The first is **time**, which includes partitioning instructions into events or cycles, basic pipelining mechanisms used to speed up instruction execution, and cycle time as a parameter for optimising program execution.
- Second, we discuss the **area**. The cost or area occupied by a particular feature is another important aspect of the architectural trade-off.
- Third, **power consumption**, affects both performance and implementation.

Design Trade - Offs

- Fourth, reliability comes into play to cope with deep submicron effects.
- Fifth, configurability provides an additional opportunity for designers to trade off recurring and nonrecurring design costs.

Requirements and Specifications

- The five basic SOC trade-offs provide a framework for analysing SOC requirements so that these can be translated into specifications.
- Cost requirements coupled with market size can be translated into the cost and process technology.
- Consider some examples:
- High-performance systems will optimise time at the expense of cost and power (and probably configurability, too).
- Low-cost systems will optimise die cost, reconfigurability, and design reuse (and perhaps low power).

Requirements and Specifications

- Wearable systems stress low power, as the power supply determines the system's weight. Since such systems, such as cell phones, frequently have real-time constraints, their performance cannot be ignored.
- Embedded systems in planes and other safety-critical applications would stress reliability, with performance and design lifetime (configurability) being important secondary considerations.
- Gaming systems would stress cost — especially production cost — and, secondarily, performance, with reliability being a lesser consideration.

1. CYCLE TIME

- The notion of time receives considerable attention from processor designers.
- It is the basic measure of performance; however, breaking actions into cycles and reducing both cycle count and cycle times are important.
- The way actions are partitioned into cycles is important. A common problem is having unanticipated “ extra ” cycles required by a basic action, such as a cache miss.
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1. CYCLE TIME

Defining a Cycle

- A cycle (of the clock) is the basic time unit for processing information.
- In a synchronous system, the clock rate is a fixed value, and the cycle time is determined by finding the maximum time to accomplish a frequent operation in the machine, such as an add or register data transfer.
- In an asynchronous system, the cycle time is simply determined by the completion of an event or operation. A completion signal is generated, allowing the next operation to begin.

1. CYCLE TIME

Optimum Pipeline

- A basic optimisation for the pipeline processor designer is the partitioning of the pipeline into concurrently operating segments.
- A greater number of segments allows a higher maximum speedup. However, each new segment incurs clocking overhead, which can adversely affect performance.

Performance

- High clock rates with small pipeline segments may or may not produce better performance.
- There are two basic factors enabling clock rate advances: (1) increased control over clock overhead and (2) an increased number of segments in the pipelines.

2. DIE AREA AND COST

- Cycle time, machine organization, and memory configuration determine machine performance. Determining performance is relatively straightforward when compared to the determination of overall cost..

Processor Area

- SOC's usually have die sizes of about 10 – 15 mm on a side.
- This die is produced in bulk from a larger wafer, perhaps 30 cm (about 12 in) in diameter.
- It might seem that one could simply increase chip size and produce fewer chips per wafer, and these larger chips could readily accommodate any function the designer might wish to include.
- Unfortunately, neither the silicon wafers nor the processing technologies are perfect.
- Defects randomly occur over the wafer surface. Large chip areas require an absence of defects over that area. If chips are too large for a particular processing technology, there will be little or no yield. Figure 2.7 illustrates yield versus chip area.

2. DIE AREA AND COST

Processor Area

- Large chip areas require an absence of defects over that area.
 - If chips are too large for a particular processing technology, there will be little or no yield.
- Figure 2.7 illustrates yield versus chip area.

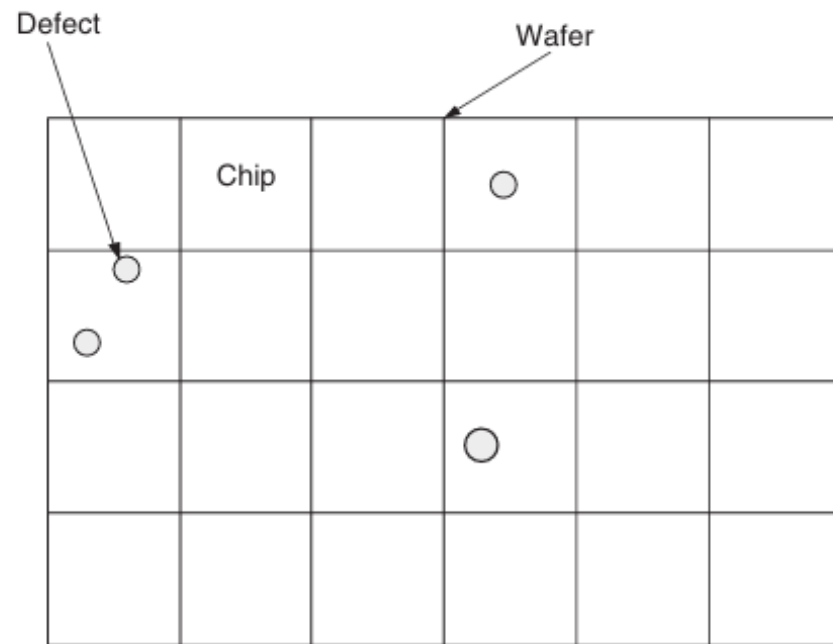


Figure 2.6 Defect distribution on a wafer.

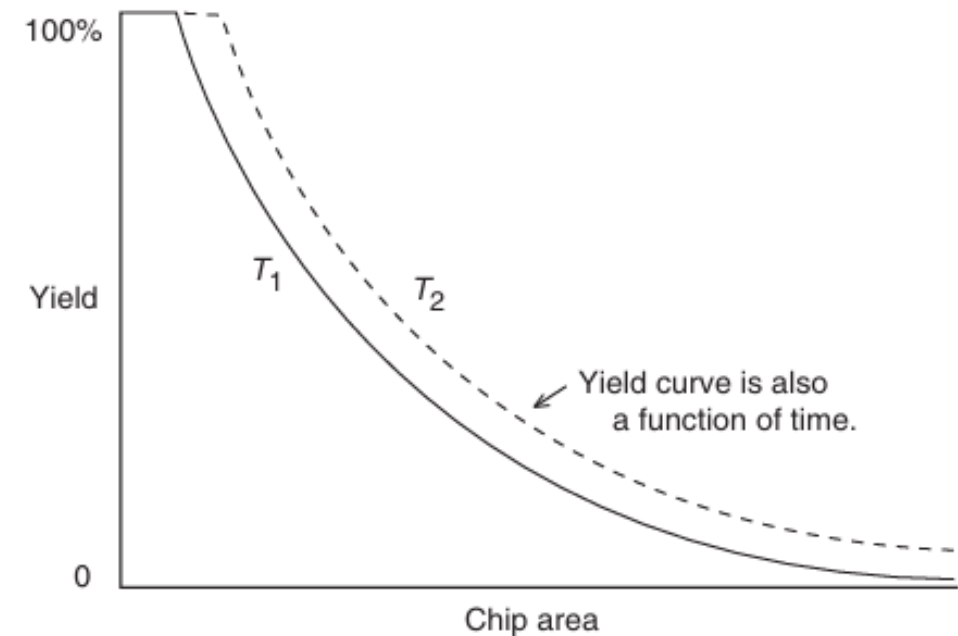
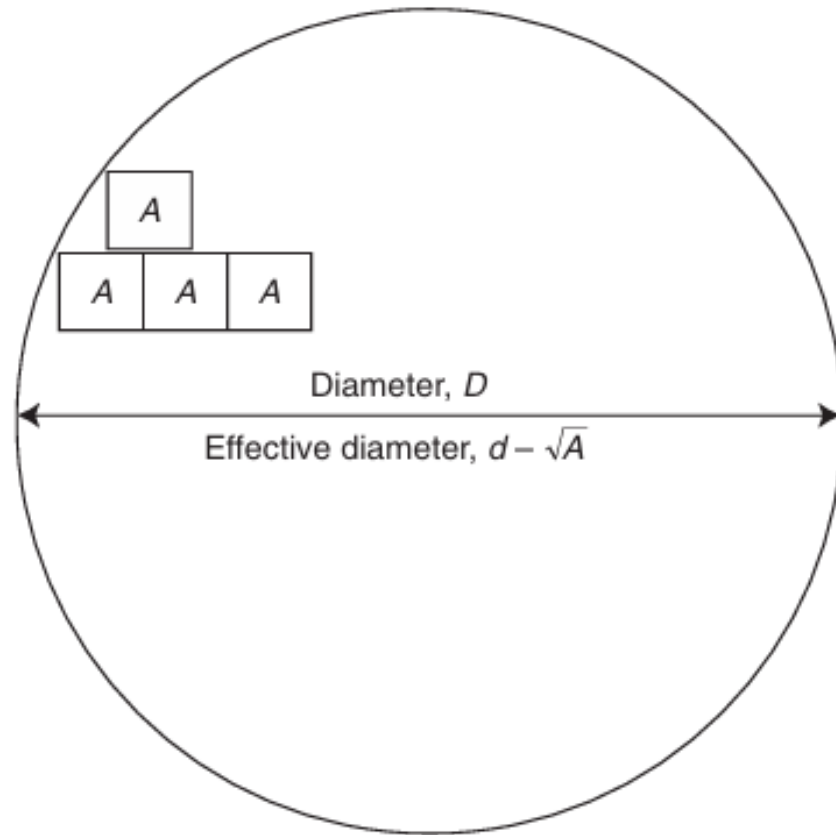


Figure 2.7 Yield versus chip area at various points in time.

2. DIE AREA AND COST

Processor Area



$$N \approx \frac{\pi}{4A} (d - \sqrt{A})^2.$$

Figure 2.8 Number of die (of area A) on a wafer of diameter d .

3. Power

- The growing demand for wireless and portable electronic appliances has recently focused attention on power consumption.
- The SIA roadmap points to increasingly higher power for microprocessor chips due to higher operating frequency, higher overall capacitance, and larger size.
- Power scales indirectly with feature size, as its primary determinant is frequency.
- At the device level, total power dissipation (P_{total}) has two major sources: dynamic or switching power and static power caused by leakage current

TABLE 2.4 Some Power Operating Environments [133]

Type	Power/Die	Source and Environment
Cooled high power	70.0W	Plug-in, chilled
High power	10.0–50.0W	Plug-in, fan
Low power	0.1–2.0W	Rechargeable battery
Very low power	1.0–100.0mW	AA batteries
Extremely low power	1.0–100.0 μ W	Button battery

4. Reliability

- The fourth important design dimension is reliability, also referred to as dependability and fault tolerance.
- As with cost and power, there are many more factors that contribute to reliability than what is done on a processor or SOC die.
- Reliability is related to die area, clock frequency, and power.
- The area increases the amount of circuitry and the probability of faults, but it also enables the use of error-correction and detection techniques.
- Higher clock frequencies increase electrical noise and noise sensitivity.
- Faster circuits are smaller and more susceptible to radiation.

4. Reliability

1. A *failure* is a deviation from a design specification.
2. An *error* is a failure that results in an incorrect signal value
3. A *fault* is an error that manifests itself as an incorrect logical result.
4. A *physical fault* is a failure caused by the environment, such as aging, radiation, temperature, or temperature cycling. The probability of physical faults increases with time.
5. A *design fault* is a failure caused by a design implementation that is inconsistent with the design specification. Usually, design faults occur early in the lifetime of a design and are reduced or eliminated over time.

5. Configurability

The motivation for adopting reconfigurable designs, primarily to manage complexity through high-performance intellectual property (IP) and avoid the risks and delays associated with fabrication.

More reasons for using reconfigurable devices, such as FPGAs, are time, area and reliability.

Time. Since FPGAs, particularly the fine-grained ones, contain an abundance of registers, they support highly pipelined designs.

Area. While it is true that the programmability of FPGAs incurs area overhead, their regularity simplifies the adoption of more aggressive manufacturing process technologies than those for application-specific integrated circuits (ASICs).

Reliability. The regularity and homogeneity of FPGAs enable the introduction of redundant cells and interconnections into their architecture. Various strategies have been developed to avoid manufacturing or runtime faults through redundant structures.

